

Victor BOUSSANGE

Applied ML Researcher, Ph.D.

🐙 github.com/vboussange 🔗 <https://victorboussange.com>
☎ +41 77 539 65 70 @ vic.boussange@gmail.com
📍 Lausanne, Switzerland 🇫🇷 French citizen
★ 227 GitHub stars earned in 860+ LinkedIn followers 📄 216 citations



Updated as of July 9, 2026.

I am an applied ML researcher working in the space of AI for (environmental) science. I develop spatio-temporal models and methods to simulate and forecast complex dynamical systems, with a current focus on biodiversity dynamics. Addressing methodological challenges in environmental science often yields innovative solutions with broader scientific applications: my work has produced original contributions in fields ranging from pure machine learning research to oncology. I am particularly interested in hybrid (physics-informed) modelling, integrating domain priors with deep learning and GPU-accelerated differentiable computing. Outside of work, I enjoy being in the mountains or at sea. Alpinism and sailing are very similar to science: success lies in making good decisions under uncertainty.

</> TECHNICAL SKILLS & OPEN-SOURCE CONTRIBUTIONS

Scientific computing	(sparse) linear algebra, (partial) differential equations, geospatial and molecular data processing pipelines, distributed computing, GPU acceleration.
ML & deep learning	neural differential equations, geospatial deep learning, graph neural networks, differentiable programming, dynamical-system & time-series forecasting, hybrid/mechanistic modelling, Bayesian inference.
Data & systems engineering	Xarray, NetCDF, Zarr, Parquet, Dask, DuckDB, Hugging Face, CI/CD pipelines.
Programming & ML frameworks	Python, Julia, SQL, Matlab, C++, Java, R, Bash; PyTorch, JAX, Optuna, Flux.jl, scikit-learn, wandb.
Open-source contributions	Member of the SciML organisation (scientific machine learning) with contributions to core packages, JAX, lineax, CUDA.jl, Flux.jl, Graphs.jl.

👛 PROFESSIONAL APPOINTMENTS

Present	Research fellow, EPFL, SCHOOL OF COMPUTER AND COMMUNICATION SCIENCES, Switzerland. Developed differentiable sparse linear solvers for Laplacian systems with application to graph learning. Currently developing GNNs for forecasting nonlocal spatio-temporal dynamics with application to disease and pest spread. Supervised 4 B.S./M.S. students in computer science.
December 2025	graph neural networks spatio-temporal forecasting team supervision
July 2027	Independent consultant, FEDERAL OFFICE FOR THE ENVIRONMENT (FOEN), Switzerland. Leading the development of GPU-accelerated, differentiable geospatial optimization software to support the design of Switzerland's ecological infrastructure.
July 2026	geospatial optimization software development graph neural networks
November 2025	Postdoctoral researcher, ETH ZÜRICH SWISS FEDERAL INSTITUTE FOR FOREST, SNOW AND LANDSCAPE (WSL), Switzerland. Built multi-scale geospatial deep learning models for biodiversity estimation, improving accuracy by 61% over state-of-the-art methods. Developed a GPU-accelerated, differentiable open-source library for ecological connectivity assessment, deployed at national scale in Switzerland. Co-supervised 2 Ph.D. students.
November 2022	geospatial deep learning graph neural networks scientific software engineering
August 2018	R&D intern, COMPAGNIE NATIONALE DU RHÔNE (CNR), France. Built optimisation software for energy trading and virtual power plant management.
March 2018	software development mathematical optimisation energy markets

🏛 EDUCATION

October 2022	Ph.D. in Applied Mathematics and Environmental Sciences, ETH ZÜRICH, Switzerland. Developed numerical methods for efficiently solving high-dimensional nonlinear PDEs, and for calibrating them against time-series data, with applications in eco-evolutionary dynamics. Advised by Prof. Dr. Loïc Pellissier, Prof. Dr. Didier Sornette, and Prof. Dr. Arnulf Jentzen.
September 2018	scientific machine learning high-dimensional PDEs biomathematics
June 2017	Master's thesis, CSIRO, Australia. Numerical continuation and bifurcation analysis applied to nonlinear dynamical systems. Advised by Dr. Thomas Poulet.
February 2017	scientific computing dynamical systems numerical methods
August 2018	B.S./M.S. in Energy and Environmental Engineering, INSA LYON, France. > Two-year intensive undergraduate course in mathematics and physics. Ranked 21st/650 students. > Three-year engineering curriculum in Energy and Environmental Systems.
September 2013	optimisation energy markets fluid mechanics thermodynamics

SELECTED SOFTWARE

AMJAX	2026
github.com/vboussange/AMJAX Algebraic multigrid solvers for sparse linear systems in JAX.	
ML library linear algebra GPU acceleration	
MUScARi	2026
github.com/vboussange/MuScaRi Implementation and pre-trained weights for MuScaRi, a model for multi-scale richness estimation.	
Model	
HYBRIDDYNAMICMODELS.JL	2025
github.com/vboussange/HybridDynamicModels.jl Training strategies for calibrating nonlinear dynamical models against time series.	
ML library time series modelling	
JAXSCAPE	2024
github.com/vboussange/jaxscape A GPU-accelerated and differentiable library for landscape genetics and ecological connectivity modelling.	
Domain-centric library graph algorithms	
HIGHDIMPDE.JL	2021
github.com/SciML/HighDimPDE.jl documentation A Julia package that breaks down the curse of dimensionality when solving nonlocal, nonlinear PDEs.	
ML library	

SELECTED PUBLICATIONS

Preprints	> Plekhanova, E., Brun, P., Fischer, J.-C., Ma, H., Boussange, V., Zimmermann, N. E., <i>Clustering biomes from space: from pixels to foundation models</i>. [SSRN] (2026). In revision at Remote Sensing of Environment. > Boussange, V., Brun, P., Malle, J. T., Midolo, G., Portier, J., Sanchez, T., Zimmermann, N. E., Axmanova, I., Bruelheide, H., Chytry, M., Kambach, S., Lososova, Z., Vecera, M., Birrun, I., Ecker, K. T., Lenoir, J., Svenning, J.C., Karger, D. N., <i>Multi-scale species richness estimation with deep learning</i>. [arXiv] (2025). Revision requested by Nature Communications.
Peer-reviewed	> Dieing, M., Volpi, M., Boussange, V., <i>Biodiversity Profile Estimation with Earth Observation</i>. Accepted at the IAPR Workshop on Pattern Recognition in Remote Sensing (PRRS 2026). > Boussange, V., Vilimelis-Aceituno, P., Pellissier, L., <i>A calibration framework to improve mechanistic forecasts with hybrid dynamic models</i>. [Methods in Ecology and Evolution] (2025). > Sapienza, F., Bolibar, J., Schäfer, F., Groenke, B., Pal, A., Boussange, V., Heimbach, P., Hooker, G., Pérez, F., Persson, P.O., Rackauckas, C., <i>Differentiable Programming for Differential Equations: A Review</i>. [arXiv] (2024), 72 pages. Accepted at SIAM Review. > Aceituno, P., Miller, J., Marti, N., Farag, Y., Boussange, V., <i>Temporal horizons in forecasting: a performance-learnability trade-off</i>. [Transactions on Machine Learning Research] (2025). > Boussange, V., Becker, S., Jentzen, A., Kuckuck, B., Pellissier, L., <i>Deep learning approximations for non-local nonlinear PDEs with Neumann boundary conditions</i>. [Partial Differential Equations and Applications], Paper no. 51, 59 pages (2023). [arXiv]

- > Skeels, A., Boschman, L. M., McFadden, I. R., Joyce, E.M., Hagen, O., Jiménez Robles, O., Bach, W., Bous-sange, V., Keggins, T., Jetz, W., Pellissier, L., *Paleoenvironments shaped the exchange of terrestrial vertebrates across Wallace's Line*.  *Science* 381, 86-92 (2023).

SUPERVISION AND TEACHING

- 2026 Semester project supervision, EPFL, IC, Switzerland. Ghali Elouahdani, "Accelerating ecological connectivity analysis with graph neural networks".
- 2026 Semester project supervision, EPFL, IC, Switzerland. Fanny Missier, "Developing an algebraic multigrid solver in JAX".
- 2025 Master's student supervision, ETH ZÜRICH, D-INF, Switzerland. Jeffrey Zweidler, "Forecasting invasive species range expansion using ecologically informed neural networks", in collaboration with the Swiss Data Science Center.
- 2025 Master's student supervision, ETH ZÜRICH, D-INF, Switzerland. Moritz Dieing, "Attention-based deep multiple instance learning for species richness prediction", in collaboration with the Swiss Data Science Center.
- 2024-2028 Ph.D. student co-supervision, ETH ZÜRICH, D-USYS, Switzerland. Capucine Lechartre, "A mechanistic approach to biome transitions across space and time".
- 2023-2024 Lecturer, Environmental Sciences Master, ETH ZÜRICH, D-USYS, Switzerland. Course 701-1679-00L, Landscape Modelling of Biodiversity: From Global Changes to Conservation.
- 2023 Lecturer, Environmental Sciences Bachelor, ETH ZÜRICH, D-USYS, Switzerland. Course 701-3001-00L, Environmental Systems Data Science. Responsible for the unit *Supervised Deep Learning: Application*.
- 2020 Lab rotation supervision, ETH ZÜRICH, D-BSSE, Switzerland. Cecilia Valenzuela Agui, MS *Computational Biology and Bioinformatics*.
- 2020 Intern supervision, POLYTECH NICE-SOPHIA, France. Nicolas Demolin, Taste of Research internship, MS *Applied Mathematics and Modeling*.

\$ FUNDING

- 2025 100,000 CHF SNSF Spark grant for *EcoDynCast: forecasting alien invasive species range dynamics with GNNs*.
- 2024 50,000 CHF A course on reproducible research, data pipelines, and scientific computing. Coordinator: Mauro Werder.
- 2024 10,000 EUR yDiv Graduate School & Postdoc Program workshop funding for a three-day event.
- 2023 10,000 CHF WSL Biodiversity Center Innovative Workshop Grant.

LANGUAGES

French ● ● ● ● ●
English ● ● ● ● ●

Spanish ● ● ● ○ ○
German ● ● ● ○ ○

HOBBIES

- > Mountain sports. Swiss Alpine Club tour leader; ski mountaineering, alpinism, rock climbing, and enduro mountain biking.
- > Sailing and expeditions. Refit a 36-foot, 40-year-old sailing boat and sailed it from Germany to Norway; long-distance ski/bicycle traverses in Norway and the Alps.
- > Video editing.  "Vagabonds du Grand Nord", official selection at [Xplore Alps Festival 2025](#)

 See full adventure CV for details.

“ REFERENCES

Dr. Michele Volpi
SWISS DATA SCIENCE CENTER
michele.volpi@sdsc.ethz.ch

Prof. Dr. Niklaus Zimmermann
SWISS FEDERAL RESEARCH INSTITUTE
niklaus.zimmermann@wsl.ch
+41 44 739 2337

Prof. Dr. Loïc Pellissier
ETH ZÜRICH
loic.pellissier@usys.ethz.ch
+41 44 632 32 03